

ECE 3337: Signals & Systems Analysis

Lecture 7: Linear Time Invariant Systems

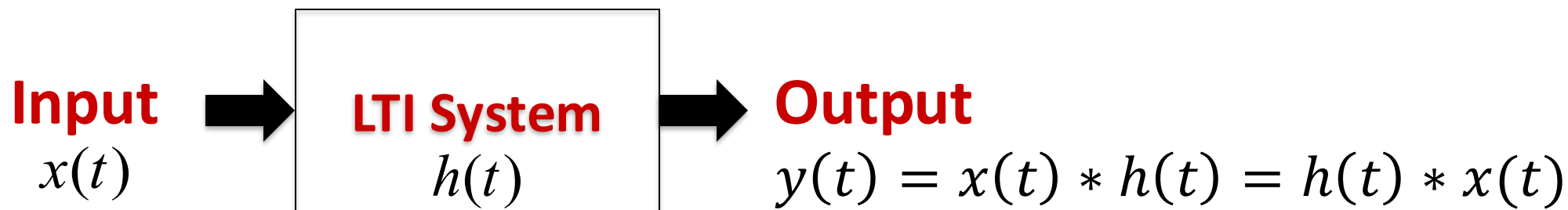
**Reference: Chapter 2
(Tables 2-1 and 2-2)**

**Please turn off your gadgets now
Grab pencil and paper, and work the equations with me**



Recap: Impulse Response of LTI System

The convolution integral allows us to calculate the response of an LTI system to *any* input, knowing *only* the impulse response!



$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau = \int_{-\infty}^{+\infty} h(\tau)x(t - \tau)d\tau$$

**MAKE SURE THAT YOU
CAN WRITE OUT THIS
INTEGRAL CORRECTLY
FROM MEMORY**

$$y(t) = u(t) \int_0^t x(\tau)h(t - \tau)d\tau$$

**SPECIAL CASE WHEN
INPUT & SYSTEM ARE
BOTH CAUSAL**



Recap: Limits and Validity

$$x(t) = f_1(t)u(t - T_1)$$

$$h(t) = f_2(t)u(t - T_2)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau) d\tau$$

$$= \int_{-\infty}^{+\infty} \underbrace{f_1(\tau)u(\tau - T_1)}_{\text{red arrow}} \underbrace{f_2(t - \tau)u(t - \tau - T_2)}_{\text{blue arrow}} d\tau$$

$$u(\tau - T_1) = 1 \text{ when } \tau > T_1$$

$$u(t - \tau - T_2) = 1 \text{ when } t - \tau - T_2 > 0$$

$$\tau < t - T_2$$



Recap: Limits and Validity

$$u(\tau - T_1) = 1 \text{ when } \tau > T_1$$

$$u(t - \tau - T_2) = 1 \text{ when } t - \tau - T_2 > 0$$

$$\square \quad \tau < t - T_2$$

$$\square \quad T_1 < \tau < t - T_2 \quad \text{SETS INTEGRATION LIMITS}$$

$$\square \quad T_1 < t - T_2 \text{ or } t > T_1 + T_2 \quad \text{DEFINES VALID VALUES OF } t$$

$$y(t) = \left[\int_{T_1}^{t-T_2} f_1(\tau) f_2(t - \tau) d\tau \right] \underline{u(t - (T_1 + T_2))}$$

THE UNIT STEP ZEROES OUT INVALID VALUES



Convolution Properties

MEMORIZE

1. Commutative

$$x(t) * h(t) = h(t) * x(t)$$

2. Associative

$$[g(t) * h(t)] * x(t) = g(t) * [h(t) * x(t)]$$

3. Distributive

$$x(t) * [h_1(t) + \cdots + h_N(t)] = x(t) * h_1(t) + \cdots + x(t) * h_N(t)$$

4. Causal * Causal = Causal

$$y(t) = u(t) \int_0^t h(\tau) x(t - \tau) d\tau$$

5. Time-shift

$$h(t - T_1) * x(t - T_2) = y(t - T_1 - T_2)$$

6. Convolution with Impulse

$$x(t) * \delta(t - T) = x(t - T)$$

7. Width

$$\text{Width of } y(t) = \text{width of } x(t) + \text{width of } h(t)$$

8. Area

$$\text{Area of } y(t) = \text{area of } x(t) \times \text{area of } h(t)$$

9. Convolution with $u(t)$

$$y(t) = x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau \quad (\text{Ideal integrator})$$

10a. Differentiation

$$\left(\frac{d^m x}{dt^m} \right) * \left(\frac{d^n h}{dt^n} \right) = \frac{d^{m+n} y}{dt^{m+n}}$$

10b. Integration

$$\int_{-\infty}^t y(\tau) d\tau = x(t) * \left[\int_{-\infty}^t h(\tau) d\tau \right] = \left[\int_{-\infty}^t x(\tau) d\tau \right] * h(t)$$



Integration by Parts

Given a product of two functions $f(t)g(t)$, we start by choosing $f(t)$ to be the term that simplifies upon differentiation. Then, use the following direct formula:

$$\int_a^b f(t)g(t)dt = f(t)G(t)\Big|_a^b - \int_a^b f'(t)G(t)dt$$

Where $G(t) = \int g(t)dt$ $G'(t) = g(t)$

(anti-derivative, or indefinite integral).

Expand the second integral if necessary.



<https://youtu.be/WWmJVmvgljk>



Same Formula in (u, v) Notation

$$\int_a^b u(x) v'(x) dx = \left[u(x) v(x) \right]_a^b - \int_a^b u'(x) v(x) dx$$

https://en.wikipedia.org/wiki/Integration_by_parts



Exponentials

$$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} \dots \simeq 1 + t$$

$$e^a \times e^b = e^{a+b}$$

$$\frac{e^a}{e^b} = e^a \times e^{-b} = e^{a-b}$$

$$\frac{d}{dt} e^{at} = a e^{at}$$

$$\int e^{at} dt = \frac{1}{a} e^{at}$$

$$\int t e^{at} dt = \frac{(at - 1)}{a^2} e^{at}$$

$$\int t^2 e^{at} dt = \frac{(a^2 t^2 - 2at + 2)}{a^3} e^{at}$$

DERIVE THESE FIRST



Example

Calculate $I = \int t e^{at} dt$



Example

Calculate: $I = \int t e^{at} dt$

Solution: We choose $f(t) = t$, and $g(t) = e^{at}$

So, $f'(t) = 1$, and $G(t) = \frac{e^{at}}{a}$

$$\begin{aligned} I &= t \cdot \frac{e^{at}}{a} - \int 1 \cdot \frac{e^{at}}{a} dt = t \cdot \frac{e^{at}}{a} - \frac{1}{a} \int e^{at} dt \\ &= t \cdot \frac{e^{at}}{a} - \frac{e^{at}}{a^2} = \frac{e^{at}}{a^2} (at - 1) \end{aligned}$$



Table 2-2: Commonly encountered convolutions.

ALL CAUSAL

1. $u(t) * u(t) = t u(t)$

2. $e^{at} u(t) * u(t) = \left(\frac{e^{at} - 1}{a} \right) u(t)$

3. $e^{at} u(t) * e^{bt} u(t) = \left[\frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$

4. $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$

5. $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a - b) t e^{at}}{(a - b)^2} u(t),$
 $a \neq b$

6. $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$

7. $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$



Derivation of #7

Use the snapshot property to derive this entry

$$x(t) = \delta(t - T_1) \text{ and } h(t) = \delta(t - T_2)$$

Intuitively, this is a cascade of two LTI systems with delays T_1 and T_2

$$\begin{aligned} \Rightarrow y(t) &= x(t) * h(t) = \int_{-\infty}^{+\infty} x(\tau) h(t - \tau) d\tau \\ &= \int_{-\infty}^{+\infty} \delta(\tau - T_1) \delta(t - \tau - T_2) d\tau = \delta(t - \tau - T_2)|_{\tau=T_1} \\ &= \delta(t - T_1 - T_2) \end{aligned}$$

Snapshot of the second term at $\tau = T_1$



How Property 7 is Helpful

This property is especially helpful when we need to convolve signals that are delayed

Example:

Convolve $x(t) = r(t - 1)$ (ramp function) and $h(t) = u(t + 3)$

$$x(t) = r(t) * \delta(t - 1)$$

$$h(t) = u(t) * \delta(t + 3)$$

$$\begin{aligned} y(t) &= x(t) * h(t) = r(t) * u(t) * \delta(t - 1) * \delta(t + 3) \\ &= r(t) * u(t) * \delta(t + 2) \end{aligned}$$

$$= \frac{1}{2} t^2 u(t) * \delta(t + 2) = \frac{1}{2} (t + 2)^2 u(t + 2)$$



Convolution Practice

- **Please hand-derive properties 1 – 6 listed in Table 2-2 in class (pencil and paper)**
 - Make sure to write out all the steps in full, showing the integration limits and validity range calculations
 - Ask questions, get answers
 - Feel free to work in teams of two if you like
 - If you run out of time, please complete the work at home
 - Detailed solutions are posted on YouTube
 - Practice until you can do all 7 of these quickly and 100% correctly
 - Submit your work on Canvas



<https://youtu.be/xAW7g3iGjf8>



Table 2-2: Commonly encountered convolutions.

1. $u(t) * u(t) = t u(t)$
2. $e^{at} u(t) * u(t) = \left(\frac{e^{at} - 1}{a} \right) u(t)$
3. $e^{at} u(t) * e^{bt} u(t) = \left[\frac{e^{at} - e^{bt}}{a - b} \right] u(t), \quad a \neq b$
4. $e^{at} u(t) * e^{at} u(t) = t e^{at} u(t)$
5. $t e^{at} u(t) * e^{bt} u(t) = \frac{e^{bt} - e^{at} + (a - b) t e^{at}}{(a - b)^2} u(t),$
 $a \neq b$
6. $t e^{at} u(t) * e^{at} u(t) = \frac{1}{2} t^2 e^{at} u(t)$
7. $\delta(t - T_1) * \delta(t - T_2) = \delta(t - T_1 - T_2)$



For the Next Class

- **Work through the textbook examples in Sections 2.3 – 2.5**
- ***Study Sections 2.6 – 2.7 and complete activities on Zybooks.***

How to Study:

- 1. Read the assigned sections to understand the basic concepts, and aim to memorize the concepts**
- 2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
- 3. Write down your questions and bring them to class**